

# From Type Theories to Algebra: Curry–Howard Correspondences for Algebraic Structures in CAL

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# 1. Starting Point: Logical Systems

We begin with **30 logical systems / type theories** already selected from EPS (the Encyclopedia of Proof Systems)

These include systems with variations in:

- Structural rules (weakening, contraction, exchange)
- Implication and residuation
- Modalities and operators
- propositional, first-order, higher-order
- intuitionistic and classical

Question:

What semantics do these logics admit?

Rather than starting from algebra, we start from **proof theory**.

## 2. Step 1: Algebraic Semantics

For each type theory:

- 1 Does it admit an algebraic semantics?
- 2 Is it algebraizable?
- 3 What variety (if any) validates it?

We compare candidate semantics with structures catalogued in the CAL database:

- Residuated lattices
- BCK/BCI-algebras
- MV-algebras
- Bilattices
- Boolean algebras with operators

Goal: Map logical systems  $\rightarrow$  algebraic varieties.

### 3. Step 2: Categorical Semantics

Given algebraic structure, investigate categorical models:

- Monoidal closed categories
- Residuated categories
- Hyperdoctrines
- Indexed and enriched categories

Central structural diagram:

Type Theory  $\longrightarrow$  Algebra  $\longrightarrow$  Category  $\longrightarrow$  Computation

We ask:

When does Curry–Howard emerge?

## 4. Role of LLMs

Large Language Models will assist by:

- Comparing proof rules across systems
- Detecting structural similarities
- Suggesting candidate algebraic axioms
- Highlighting inconsistencies or gaps
- Clustering systems by semantic behavior

LLMs are not proving theorems.

They act as:

- Structural pattern detectors
- Hypothesis generators
- Meta-level assistants

This creates a semi-automated discovery loop.

## 5. Big Picture

Long-term aim:

A systematic theory of when  
 $\text{Algebra} \leftrightarrow \text{Logic} \leftrightarrow \text{Computation}$

Deliverables:

- A semantic classification of the 30 logics
- Criteria for existence of Curry–Howard correspondences
- New term calculi where appropriate
- Publications bridging algebraic logic, category theory, and AI

Methodological principle:

*Start from syntax. Recover semantics.*